

Diarrhea and Dehydration in Children: Texts

Text A

Definition and Classification of Diarrhea in Children

Diarrhea is defined as the passage of three or more loose or watery stools within a 24-hour period. In children, it is one of the leading causes of morbidity and mortality worldwide, particularly in low- and middle-income countries, where inadequate access to healthcare and clean water amplifies its impact. The World Health Organization estimates that diarrheal disease contributes to nearly half a million deaths in children under five years of age annually.

Diarrhea in children is classified according to its duration. Acute diarrhea lasts fewer than 14 days and is the most commonly encountered form in clinical practice. Persistent diarrhea is defined as an episode lasting 14 days or more but fewer than 30 days and carries an increased risk of nutritional deterioration. Chronic diarrhea refers to episodes extending beyond 30 days and often indicates an underlying gastrointestinal condition requiring further investigation.

Infectious causes predominate in paediatric populations and include viral pathogens such as rotavirus, norovirus, and adenovirus; bacterial organisms including *Salmonella*, *Campylobacter*, *Shigella*, and enterotoxigenic *Escherichia coli*; and parasitic infections such as *Giardia lamblia*. Rotavirus remains the most common viral cause globally and accounts for a significant proportion of hospitalisations in young children.

Dehydration is the most serious complication of diarrhea and is classified as mild, moderate, or severe based on clinical signs. Mild dehydration presents with thirst and minimal clinical findings, while moderate dehydration is characterised by sunken eyes, dry mucous membranes, and reduced skin turgor. Severe dehydration involves altered consciousness and signs of circulatory compromise.

Continued fluid and nutritional loss without adequate replacement may lead to hypovolemic shock and, if untreated, death. Electrolyte disturbances including hyponatraemia and metabolic acidosis may further complicate the clinical course.

Text B

Risk Factors and Prevention Strategies

Several patient-related and environmental factors increase a child's susceptibility to diarrheal illness and the risk of severe dehydration.

Age is a key determinant, with children under two years of age being at greatest risk due to immature immune function and dependence on caregivers for hygiene practices. Malnutrition impairs intestinal mucosal integrity and immune defences, increasing both the frequency and severity of diarrheal episodes. Zinc deficiency in particular compromises mucosal repair mechanisms and cellular immune responses.

Environmental risk factors include inadequate access to clean water, poor sanitation, and unhygienic food preparation practices. Lack of breastfeeding in infancy removes a key source of passive immunity, particularly secretory immunoglobulin A, and increases the child's exposure to contaminated feeds and utensils.

Preventive strategies aim to reduce pathogen exposure and strengthen host resistance. Exclusive breastfeeding for the first six months of life is strongly recommended and significantly reduces diarrheal morbidity. Rotavirus vaccination has demonstrated effectiveness in reducing the incidence of severe rotavirus diarrhea in children and is now incorporated into national immunisation programmes in many countries. Promotion of hand hygiene with soap and water and provision of safe water are fundamental public health measures.

Oral rehydration solution (ORS) administration at the onset of diarrhea prevents progression to significant dehydration. Zinc supplementation given alongside ORS has been shown to reduce the duration and severity of acute diarrheal episodes and is recommended by the World Health Organization for all children presenting with acute diarrhea.

Text C

Clinical Assessment of Dehydration in Children

Clinical Feature	Possible Indication	Recommended Action
Sunken eyes and dry mucous membranes	Moderate dehydration	Initiate oral rehydration therapy
Reduced skin turgor on pinching	Moderate to severe dehydration	Assess fluid deficit and increase fluid intake
Altered consciousness or lethargy	Severe dehydration	Commence intravenous fluid resuscitation
Bloody diarrhea with fever	Bacterial (invasive) infection	Obtain stool culture and initiate appropriate antibiotics
Vomiting preventing adequate oral rehydration	Risk of aspiration and worsening dehydration	Consider nasogastric rehydration or intravenous fluids
Persistent diarrhea exceeding 14 days	Risk of malnutrition and micronutrient deficiency	Conduct nutritional assessment and refer to specialist

Prompt clinical assessment and targeted management are essential to prevent progression to hypovolemic shock, multi-organ failure, or death. All findings should be interpreted alongside the child's overall clinical presentation and nutritional status.

Text D

Step-by-Step Management of Diarrhea and Dehydration

Step 1: Initial clinical assessment

All children presenting with diarrhea should undergo a thorough history and physical examination, including assessment of stool frequency, consistency, presence of blood or mucus, vomiting, fever, and recent feeding history. Vital signs, body weight, and a detailed evaluation of hydration status should be documented at presentation.

Step 2: Classification of dehydration

Clinicians should evaluate skin turgor, mucous membrane moisture, eye appearance, level of consciousness, and capillary refill time to determine whether dehydration is absent, mild to moderate, or severe. In infants, fontanelle tension and the presence of tears provide additional diagnostic information.

Step 3: Rehydration plan

Children with no dehydration receive increased fluids at home using ORS (Plan A). Moderate dehydration requires supervised ORS of 75 ml per kilogram over four hours in a healthcare facility (Plan B). Severe dehydration requires prompt intravenous fluid resuscitation with Ringer's lactate or normal saline at 100 ml per kilogram (Plan C), with continuous monitoring.

Step 4: Nutritional support

Feeding should not be withheld during diarrhea, as continued feeding maintains intestinal integrity, supports recovery, and prevents malnutrition. Age-appropriate foods should be resumed as soon as the child can tolerate oral intake, and zinc supplementation is recommended for a minimum of ten days.

Step 5: Monitoring and discharge planning

Children should be monitored for resolution of dehydration, reduction in stool frequency, and return of normal feeding. Caregivers should be educated on recognising warning signs and when to seek further medical attention before the child is discharged.